Test Plan Documentation

[GitHub]: https://github.com/DanielJ0131/interactive-house/tree/main/SG3_Devices/SG3_Artifacts

Subgroup 3: Devices

Revision History

Date	Version	Description	Author
08/May/26	1.0	Updated file to contain Table of content and figures.	Mustafa Al-Bayati, Sergej Macut, Tony Nguyen, Ryad Hazin
08/May/26	1.1	Initial draft of the test plan, defining unit testing and documentation scope for the ARDUINO_REFACTOR project.	Mustafa Al-Bayati, Sergej Macut, Tony Nguyen, Ryad Hazin
08/May/26	1.2	Added PlatformIO native unit testing setup and Unity framework configuration.	Sergej Macut Ryad Hazin Tony Nguyen
08/May/26	1.3	Added device unit tests for BuzzerDevice, Fan, DoorDevice, and WindowDevice.	Mustafa Al-Bayati

Test Plan

Task Delegation

Here is a balanced breakdown of the unit testing and documentation workload, divided by the folders and files shown in your `src` directory:

- **Ryad** will be responsible for the core logic, configuration, and system safety. This includes writing unit tests and documentation for the `core/` folder (`CommandHandler.cpp`, `CommandHandler.h`, `Observer.h`, `Subject.h`), the `config/` folder (`config.cpp`, `config.h`), and the `safety/` folder (`GasSafety.cpp`, `GasSafety.h`) Figure[3].

- **Mustafa** will handle the hardware device representations. Due to the size of the folder, Mustafa will focus entirely on the `devices/` folder, covering tests and documentation for `BuzzerDevice`, `Devices` (base/manager), `DoorDevice`, `Fan`, and `WindowDevice` Figure[4] and [5].

- **Sergej** will take charge of the system inputs and safety mechanisms. This involves unit testing and documenting the `sensors/` folder (`sensors.cpp`, `sensors.h`, `SteamSensor.cpp`, `SteamSensor.h`), the `music/` folder (`MusicEngine.cpp`, `MusicEngine.h`), and the entry point file (`main.cpp`) Figure[2].

- **Tony** will manage the outputs, media, and the main execution loop. Tony will write tests and documentation for the `display/` folder (`LCDManager.cpp`, `LCDManager.h`), and the `gateway.js` file.

Figures

```
================================================================= SUMMARY =================================================================
Environment    Test          Status    Duration
-------------  ------------  --------  ------------
uno            test_music    SKIPPED
uno            test_sensors  SKIPPED
native         test_music    PASSED    00:00:01.383
native         test_sensors  PASSED    00:00:01.426
================================================= 8 test cases: 8 succeeded in 00:00:02.809 =================================================
```

Figure 2: Music and Sensors tests

```
================================================================= SUMMARY =================================================================
Environment    Test         Status    Duration
-------------  -----------  --------  ------------
uno            test_config  PASSED    00:00:09.332
uno            test_core    PASSED    00:00:10.646
uno            test_safety  PASSED    00:00:09.166
================================================ 10 test cases: 10 succeeded in 00:00:29.144 ================================================
```

Figure 3; Config, Core and Safety tests

```
musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$ python -m platformio test -e native
Verbosity level can be increased via `-v, -vv, or -vvv` option
Collected 1 tests

Processing test_devices in native environment
------------------------------------------------------------------------------------------------------------------------------
Building...
Testing...
test\test_devices\test_main.cpp:18: test_buzzer [PASSED]
---------------------------------------------------- native:test_devices [PASSED] Took 1.06 seconds ----------------------------------------------------

================================================================= SUMMARY =================================================================
Environment    Test          Status    Duration
-------------  ------------  --------  ------------
native         test_devices  PASSED    00:00:01.062
====================================================== 1 test cases: 1 succeeded in 00:00:01.062 ======================================================

musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$
```

Figure 4; Devices testing

```
musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$ python -m platformio run -e native -t clean
native         SUCCESS   00:00:00.555
============================================================ 1 succeeded in 00:00:00.555 ============================================================

musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$ python -m platformio test -e native
Verbosity level can be increased via `-v, -vv, or -vvv` option
Collected 1 tests

Processing test_devices in native environment
------------------------------------------------------------------------------------------------------------------------------
Building...
Testing...
test\test_devices\test_main.cpp:77: test_buzzer_turns_on        [PASSED]
test\test_devices\test_main.cpp:78: test_buzzer_turns_off       [PASSED]
test\test_devices\test_main.cpp:80: test_fan_turns_correct_direction     [PASSED]
test\test_devices\test_main.cpp:82: test_door_closes     [PASSED]
test\test_devices\test_main.cpp:84: test_window_closes   [PASSED]
---------------------------------------------------- native:test_devices [PASSED] Took 1.27 seconds ----------------------------------------------------

================================================================= SUMMARY =================================================================
Environment    Test          Status    Duration
-------------  ------------  --------  ------------
native         test_devices  PASSED    00:00:01.273
====================================================== 5 test cases: 5 succeeded in 00:00:01.273 ======================================================
```

Figure 5; Buzzer, Door, Fan and Window

```
esting -vvv
Flash: [=         ]  11.7% (used 3774 bytes from 32256 bytes)
.pio\build\uno\firmware.elf  :

section                      size      addr

.data                         370   8388864

.text                        3404         0

.bss                          223   8389234

.comment                       17         0

.note.gnu.avr.deviceinfo       64         0

.debug_aranges                256         0

.debug_info                  3020         0

.debug_abbrev                1602         0

.debug_line                  1290         0

.debug_str                    520         0

Total                       10766



--------------------------------- uno:* [PASSED] Took 1.38 seconds ---------------------------------

============================================= SUMMARY =============================================
Environment    Test    Status    Duration
-------------  ------  --------  ------------
uno            *       SKIPPED   00:00:01.385
native         *       SKIPPED
============================ 0 test cases: 0 succeeded in 00:00:01.385 ============================
```

```
PASS  test/gateway/gateway.test.js
  Gateway.js - Full Logic Suite
    Basic Helpers
      √ normalize: should trim and lowercase (4 ms)
      √ normalizeBoolean: should recognize 'on'/'off' strings
      √ clampInt: should respect boundaries
    Music Logic
      √ parseNotes: should parse semicolon separated string (2 ms)
      √ serializeNotes: should format for Arduino transmission (1 ms)
      √ getMusicState: should extract play and notes from complex data (1 ms)
    State Management
      √ getFirestoreCommandState: should map device states correctly (2 ms)
      √ hasChanges: should detect differences between objects (2 ms)

Test Suites: 1 passed, 1 total
Tests:       8 passed, 8 total
Snapshots:   0 total
Time:        0.59 s, estimated 1 s
Ran all test suites.
```

Figure 7. Gateway test